

EXPERIMENT NO 14

Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

Aim:

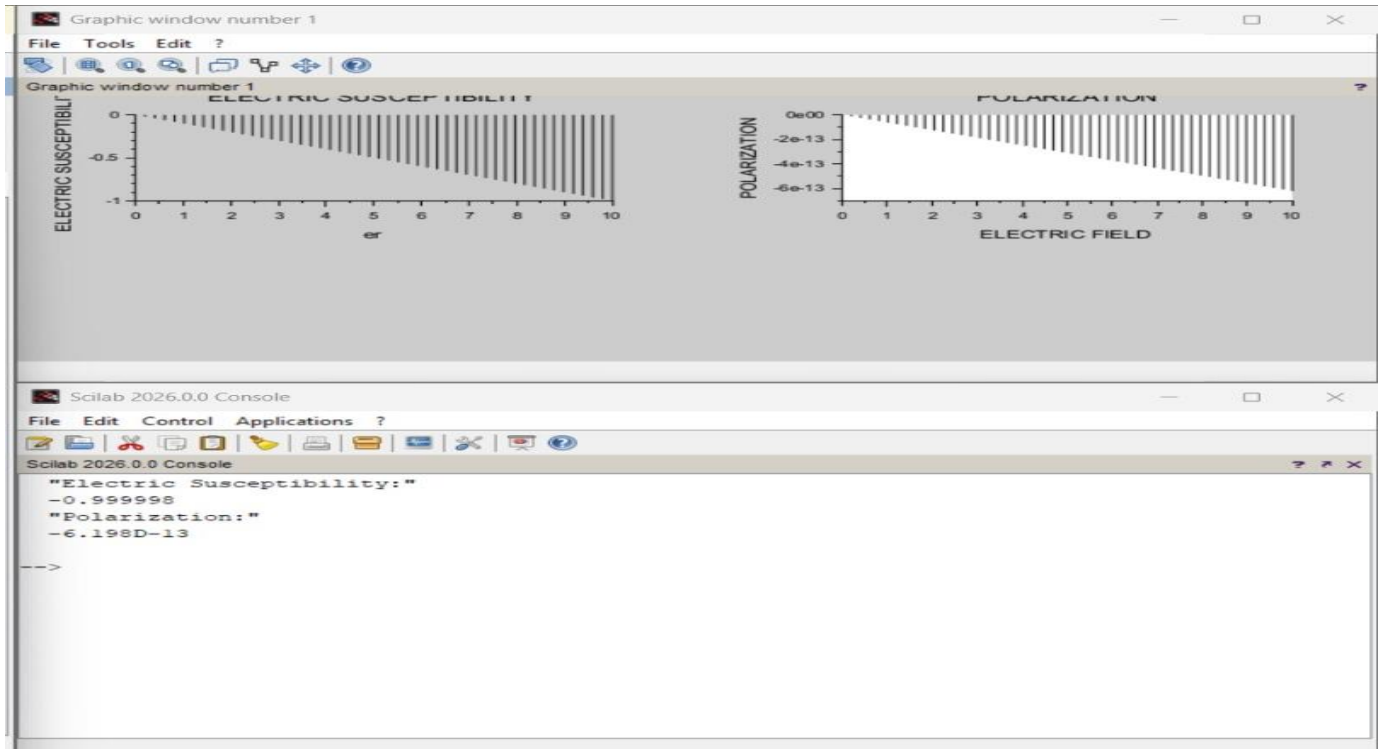
To determine and plot polarization in different dielectric medium with their relative permittivity and to find electric susceptibility.

Software required:

- SCILAB version 6.0.1

Program:

Output:



Theoretical calculations:

Exp 1.14

$$\epsilon_r = 2 \times 10^{-6}$$

$$\text{Electric field} = 7 \times 10^8$$

$$\chi_e = \epsilon_r - 1$$

$$\chi_e = 2 \times 10^{-6} - 1$$

$$= -0.99998$$

polarization

$$P = \chi_e \epsilon_0 E$$

$$P = (-0.99998) (8.854 \times 10^{-12}) (7 \times 10^8)$$

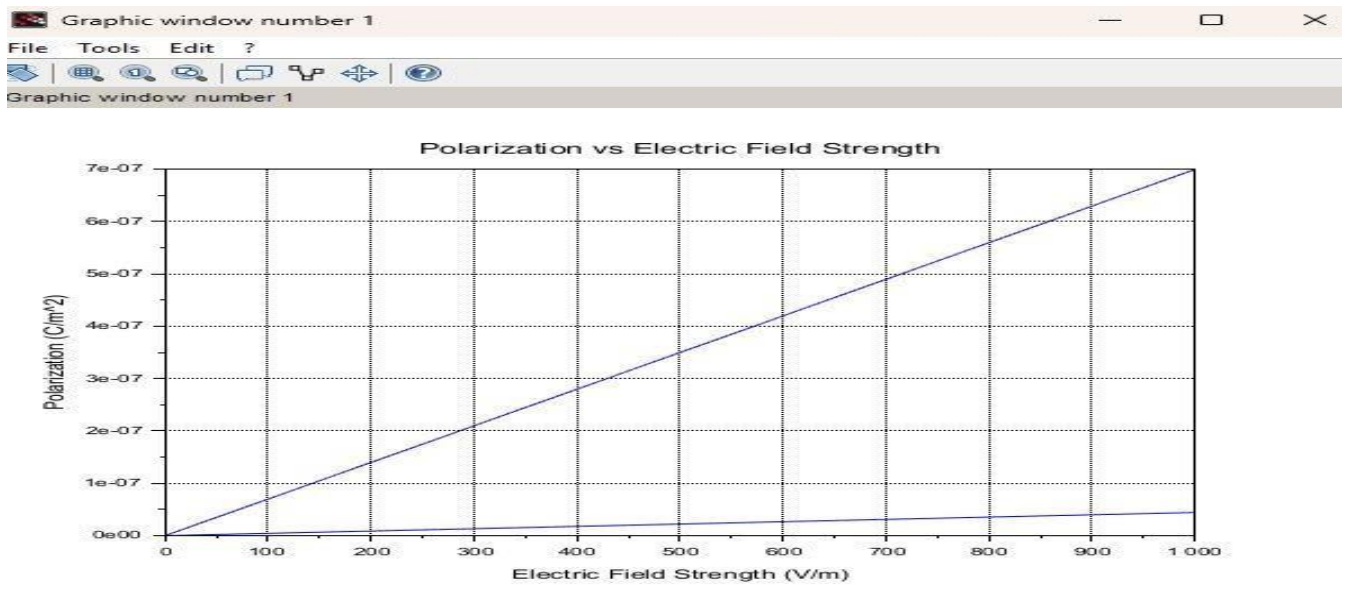
$$P = -6.19 \times 10^{-3} \text{ C/m}^2$$

Case 1: To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci
1 clear
2 clear
3
4 // Permittivity of free space
5 epsilon_0 = 8.854e-12;
6
7 // Electric field strength
8 E = linspace(0, 1000, 100);
9
10 // Relative permittivity
11 er_air = 1;
12 er_water = 80;
13 er_glass = 4;
14
15 // Electric susceptibility
16 chi_air = er_air - 1;
17 chi_water = er_water - 1;
18 chi_glass = er_glass - 1;
19
20 // Polarisation
21 P_air = epsilon_0 * chi_air * E;
22 P_water = epsilon_0 * chi_water * E;
23 P_glass = epsilon_0 * chi_glass * E;
24
25 // Create graph
26 clf();
27 clf();
28
29 // Plot Air
30 plot(E, P_air);
31
32 // Plot Water
33 plot(E, P_water);
34
35 // Plot Glass
36 plot(E, P_glass);
37
38 // Labels
39 xlabel('Electric Field Strength (V/m)');
40 ylabel('Polarisation (C/m^2)');
41 title('Polarisation vs Electric Field Strength');
42
43 // Grid
44 grid();
45
```

Output:

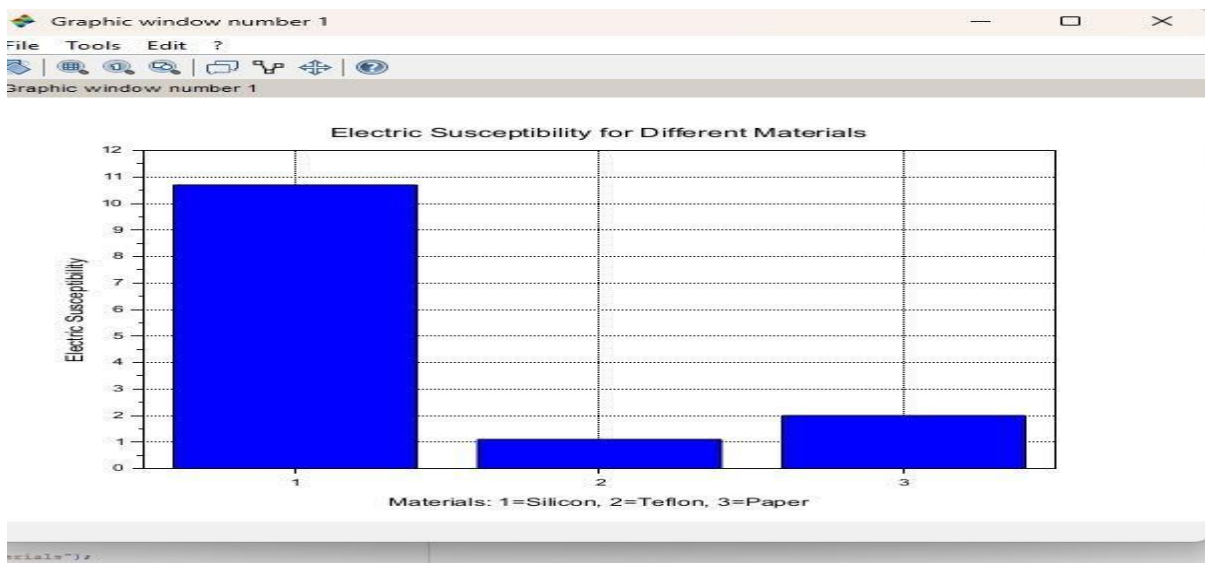


Case 2: To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci
1 clear;
2 clear;
3
4 // Constant
5 epsilon0 = 8.854e-12;
6
7 // Relative permittivity
8 epsilon_r = [11.7 2.1 3.0];
9
10 // Electric susceptibility
11 chi_e = epsilon_r - 1;
12
13 // Electric field
14 E = 1e6;
15
16 // Polarisation
17 polarisation = epsilon0 .* chi_e .* E;
18
19 // Display values
20
21 disp("Electric Susceptibility:");
22 disp(chi_e);
23
24 disp("Polarisation -(C/m^2):");
25 disp(polarisation);
26
27 // Graph 1:- Electric Susceptibility
28
29 clf();
30
31 bar(chi_e);
32
33 xlabel("Materials: 1=Silicon, 2=Teflon, 3=Paper");
34 ylabel("Electric Susceptibility");
35 title("Electric Susceptibility for Different Materials");
36 grid();
37
38 // Graph 2:- Polarisation
39
40 clf();
41
42 bar(polarisation);
43
44 xlabel("Materials: 1=Silicon, 2=Teflon, 3=Paper");
45 ylabel("Polarisation -(C/m^2)");
46 title("Polarisation for Different Materials");
47 grid();
48
```

Output:



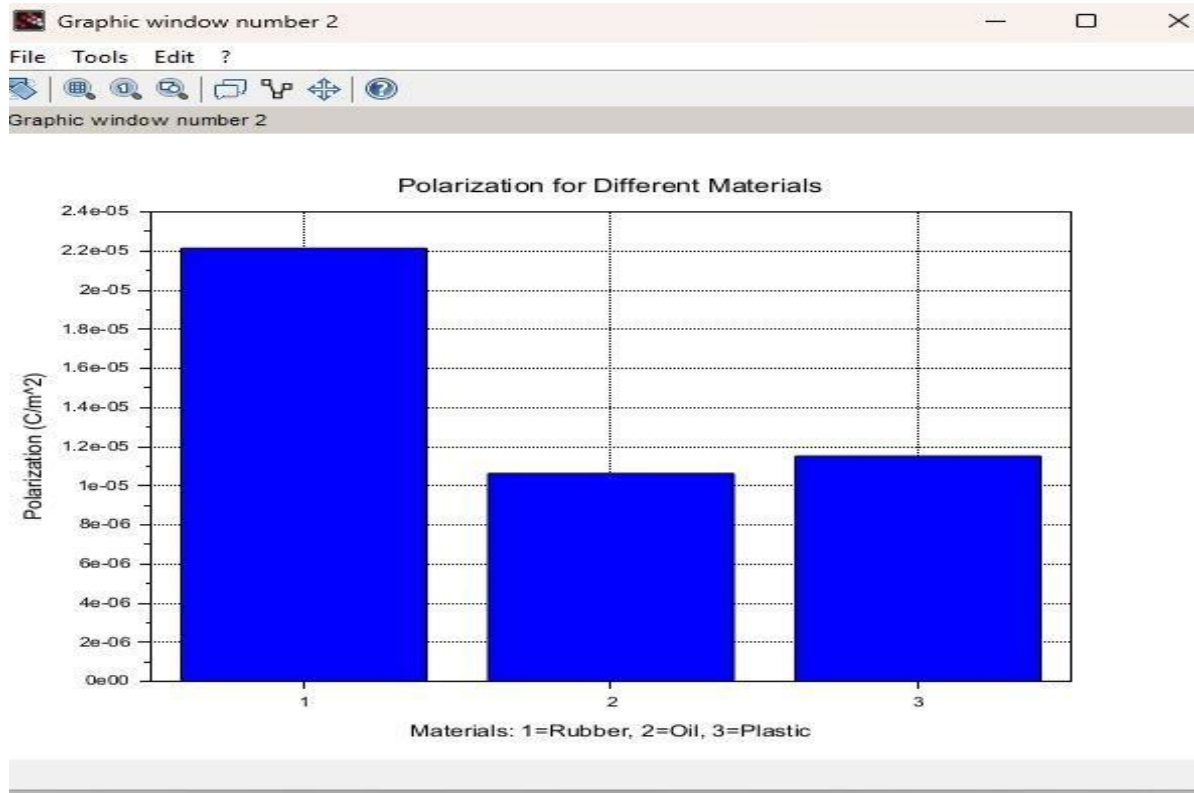
Case 3: To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

1 clc;
2 clear;
3
4 // Constant
5 epsilon0 = 8.854e-12;
6
7 // Relative permittivity
8 epsilon_r = [2.5 2.2 2.2];
9
10 // Electric susceptibility
11 chi_e = epsilon_r - 1;
12
13 // Applied electric field
14 E = 1e6;
15
16 // Polarisation
17 polarisation = epsilon0 .* chi_e .* E;
18
19
20 // =====
21 // DISPLAY RESULTS
22 // =====
23
24 disp("Electric Susceptibility:");
25 disp(chi_e);
26
27 disp("Polarisation (C/m^2):");
28 disp(polarisation);
29
30 // =====
31 // GRAPH 1: ELECTRIC SUSCEPTIBILITY
32 // =====
33
34 clf(1);
35
36 clf();
37
38 bar(chi_e);
39
40 xlabel("Materials: -1=Rubber, -2=Oil, -3=Plastic");
41 ylabel("Electric Susceptibility");
42 title("Electric Susceptibility for Different Materials");
43
44
45
46 // =====
47 // GRAPH 2: POLARISATION
48 // =====
49
50
51 clf(2);
52
53 clf();
54
55 bar(polarisation);
56
57 xlabel("Materials: -1=Rubber, -2=Oil, -3=Plastic");
58 ylabel("Polarisation (C/m^2)");
59 title("Polarisation for Different Materials");
60
61 xgrid();
62
```

Output:



Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B) for the following conditions

- ☐ The material is ferromagnetic with relative permeability (μ_r) of 500.
- ☐ The magnetic field intensity (H) is held constant at 100 A/m.
- ☐ The length of the material varies from 0.1 m to 2 m.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1 clear;
2 clear;
3 clf;
4
5 // Constants
6 mu_r = 500; // Relative permeability
7 H = 100; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^-7; // Permeability of free space (H/m)
9
10 // Lengths from 0.1 m to 2 m
11 lengths = 0.1:0.1:2;
12
13 // Initialise array for magnetic flux density
14 B_values = zeros(1, length(lengths));
15
16 // Calculate magnetic flux density
17 for i = 1:length(lengths)
18     B_values(i) = mu_r * mu_0 * H;
19 end
20
21 // Display results
22 disp('Length (m) ----- Magnetic Flux Density (T)');
23
24 for i = 1:length(lengths)
25     fprintf('%15.6f\n', lengths(i), B_values(i));
26 end
27
28 // Plot the results
29 plot(lengths, B_values, '-o');
30 xlabel('Length of Ferromagnetic Material (m)');
31 ylabel('Magnetic Flux Density B (T)');
32 title('Effect of Length on Magnetic Flux Density');
33 grid();
34
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Documents

My Music

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EXP 14.sci

EXP 14.sci

Scilab 2026.1.0 Console

"Length (m) Magnetic Flux Density (T)"

0.1	0.062832
0.2	0.062832
0.3	0.062832
0.4	0.062832
0.5	0.062832
0.6	0.062832
0.7	0.062832
0.8	0.062832
0.9	0.062832
1.0	0.062832
1.1	0.062832
1.2	0.062832
1.3	0.062832
1.4	0.062832
1.5	0.062832
1.6	0.062832
1.7	0.062832
1.8	0.062832
1.9	0.062832
2.0	0.062832

-->

Variable Browser

Name	Value	Type	Visibility	Memory
B_values	1x20	Double	local	368 B
H	100	Double	local	216 B
i	20	Double	local	216 B
lengths	1x20	Double	local	368 B
mu_0	1.26e-06	Double	local	216 B
mu_r	500	Double	local	216 B

Command History

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci X

```
1 clear;
2 clear;
3 clf;
4
5 // Constants
6 mu_r = 500; // Relative permeability
7 H = 100; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^-7; // Permeability of free space (H/m)
9
10 // Lengths from 0.1 m to 2 m
11 lengths = 0.1:0.1:2;
12
13 // Initialise array for magnetic flux density
14 B_values = zeros(1, length(lengths));
15
16 // Calculate magnetic flux density
17 for i = 1:length(lengths)
18     B_values(i) = mu_r * mu_0 * H;
19 end
20
21 // Display results
22 disp('Length (m) ----- Magnetic Flux Density (T)');
23
24 for i = 1:length(lengths)
25     fprintf('%15.6f\n', lengths(i), B_values(i));
26 end
27
28 // Plot the results
29 plot(lengths, B_values, '-o');
30 xlabel('Length of Ferromagnetic Material (m)');
31 ylabel('Magnetic Flux Density B (T)');
32 title('Effect of Length on Magnetic Flux Density');
33 grid();
34
```

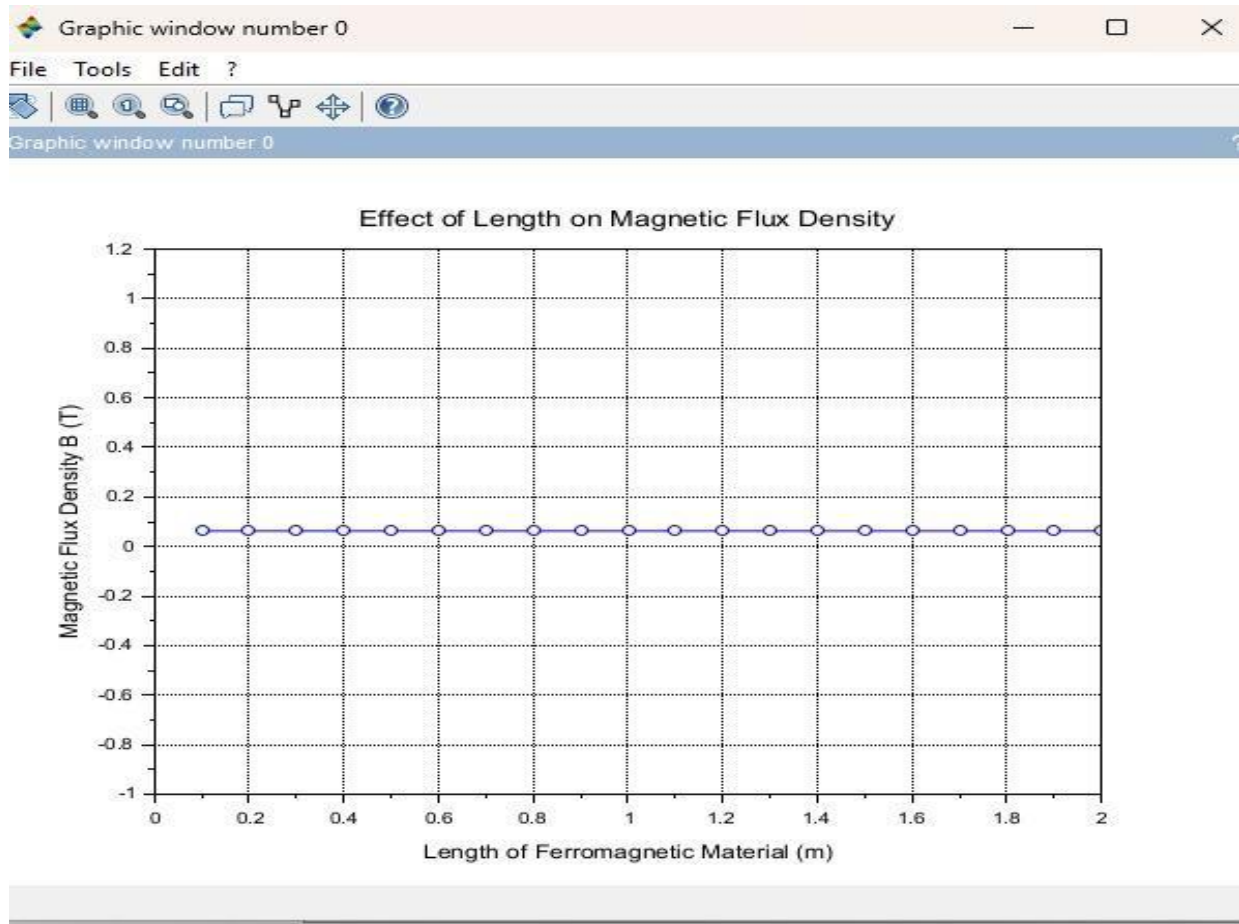
Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

11:29 AM
8/30/2026



Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity.

Conditions:

- Material: Soft iron (same as in Case Study 1)
- Length (L): Constant at 10 cm
- Magnetic Field Intensity (H): Constant at 1000 A/m
- Width (W): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
[Icons]
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x
1  clc;
2  clear;
3  clf;
4
5  // Constants
6  L = 0.1; // Length in meters (10 cm)
7  H = 1000; // Magnetic field intensity (A/m)
8  mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
9
10 // Widths in meters (2 cm to 10 cm)
11 widths = [0.02, 0.04, 0.06, 0.08, 0.10];
12 num_widths = length(widths);
13
14 // Relative permeability of soft iron
15 mu_r = 1000;
16
17 // Initialise array
18 B_values = zeros(1, num_widths);
19
20 // Calculate magnetic flux density
21 for i = 1:num_widths
22     B_values(i) = mu_0 * mu_r * H;
23 end
24
25 // Display results
26 disp("Width (m) ----- Flux Density (T)");
27
28 for i = 1:num_widths
29     fprintf("%.2f-----%.6f\n", widths(i), B_values(i));
30 end
31
32 // Plot the results
33 plot(widths, B_values, '-o');
34
35 xlabel("Width of Ferromagnetic Material (m)");
36 ylabel("Magnetic Flux Density B (T)");
37 title("Effect of Width on Magnetic Flux Density");
38
39 xgrid();
40
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Documents

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ex 6.sci

EXP 14.sci

Scilab 2026.1.0 Console

"Width (m) Flux Density (T)"

```
0.02 1.256637
0.04 1.256637
0.06 1.256637
0.08 1.256637
0.10 1.256637

--> |
```

Variable Browser

Name	Value	Type	Visibility	Memory
B_values	1x5	Double	local	248 B
H	1e+03	Double	local	216 B
L	0.1	Double	local	216 B
i	5	Double	local	216 B
mu_0	1.26e-06	Double	local	216 B
mu_r	1e+03	Double	local	216 B
num_widths	5	Double	local	216 B
widths	1x5	Double	local	248 B

Command History

```
-4
-2
-1
-3
-4
-5
```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci x

```
1  clc;
2  clear;
3  clf;
4
5  // Constants
6  L = 0.1; // Length in meters (10 cm)
7  H = 1000; // Magnetic field intensity (A/m)
8  mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
9
```

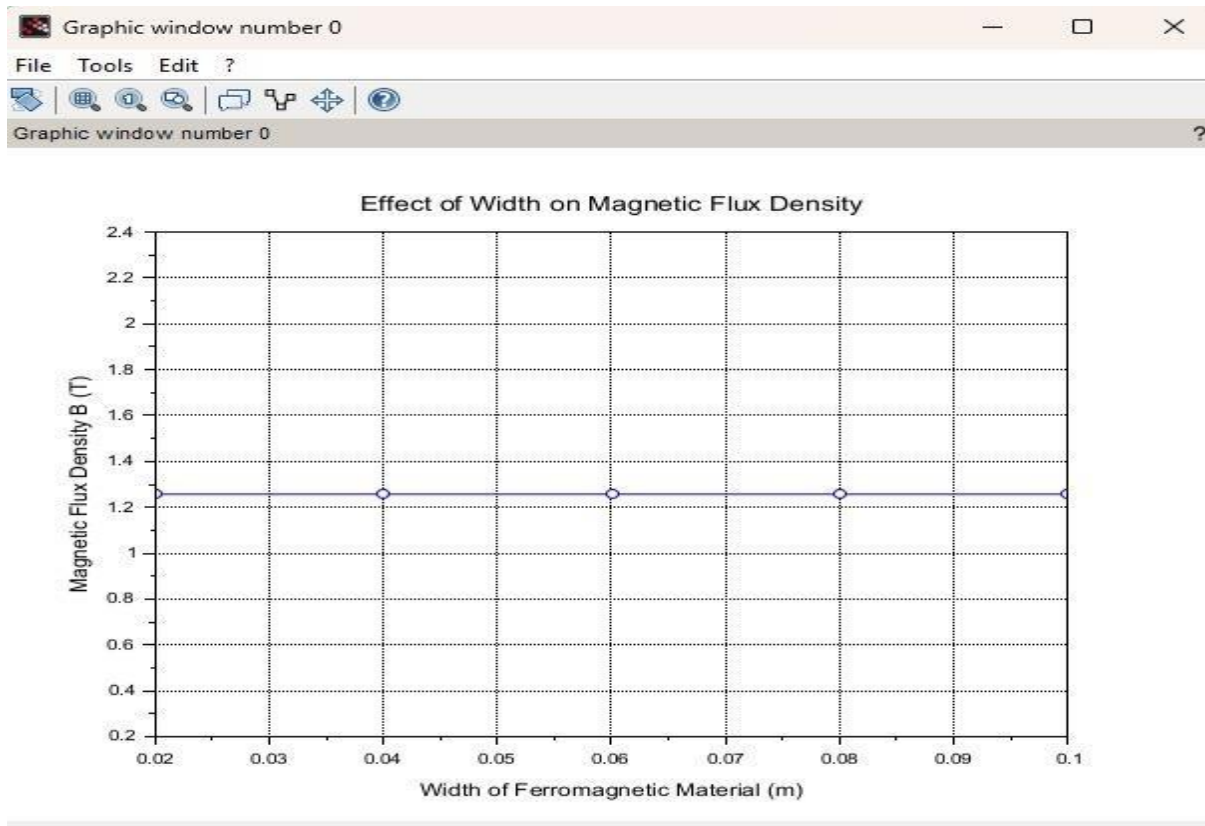
Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

11:31 AM 8/30/2026



Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material changes with different values of magnetic field intensity at constant length and width.

Conditions:

- Material: Soft iron (same as in the previous studies)
- Length (L): Constant at 10 cm
- Width (W): Constant at 5 cm
- Magnetic Field Intensity (H): Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m)

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

1 clear;
2 clear;
3 clf();
4
5 // Define constants
6 L = 0.1; // Length in meters (10 cm)
7 W = 0.05; // Width in meters (5 cm)
8
9 H_values = [200, 500, 1000, 1500, 2000]; // H in A/m
10
11 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
12 mu_r = 1000; // Relative permeability of soft iron
13
14 // Initialize array for magnetic flux density
15 B_values = zeros(1, length(H_values));
16
17 // Calculate B for each H value
18 for i = 1:length(H_values)
19     H = H_values(i);
20     mu = mu_0 * mu_r; // Absolute permeability
21     B_values(i) = mu * H; // B = mu * H
22 end
23
24 // Display results
25 disp('Magnetic Field Intensity (A/m) --- Magnetic Flux Density (T)');
26
27 for i = 1:length(H_values)
28     fprintf('%10.0f-----%.6f\n', H_values(i), B_values(i));
29 end
30
31 // Plot the results
32 plot(H_values, B_values, '-o');
33
34 xlabel('Magnetic Field Intensity H (A/m)');
35 ylabel('Magnetic Flux Density B (T)');
36 title('Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron');
37
38 grid();
39
```

Output:

The screenshot shows the SciLab 2026.1.0 Console window. The main window displays the output of the script, which is a table of magnetic field intensity (H) and magnetic flux density (B) values. The output is as follows:

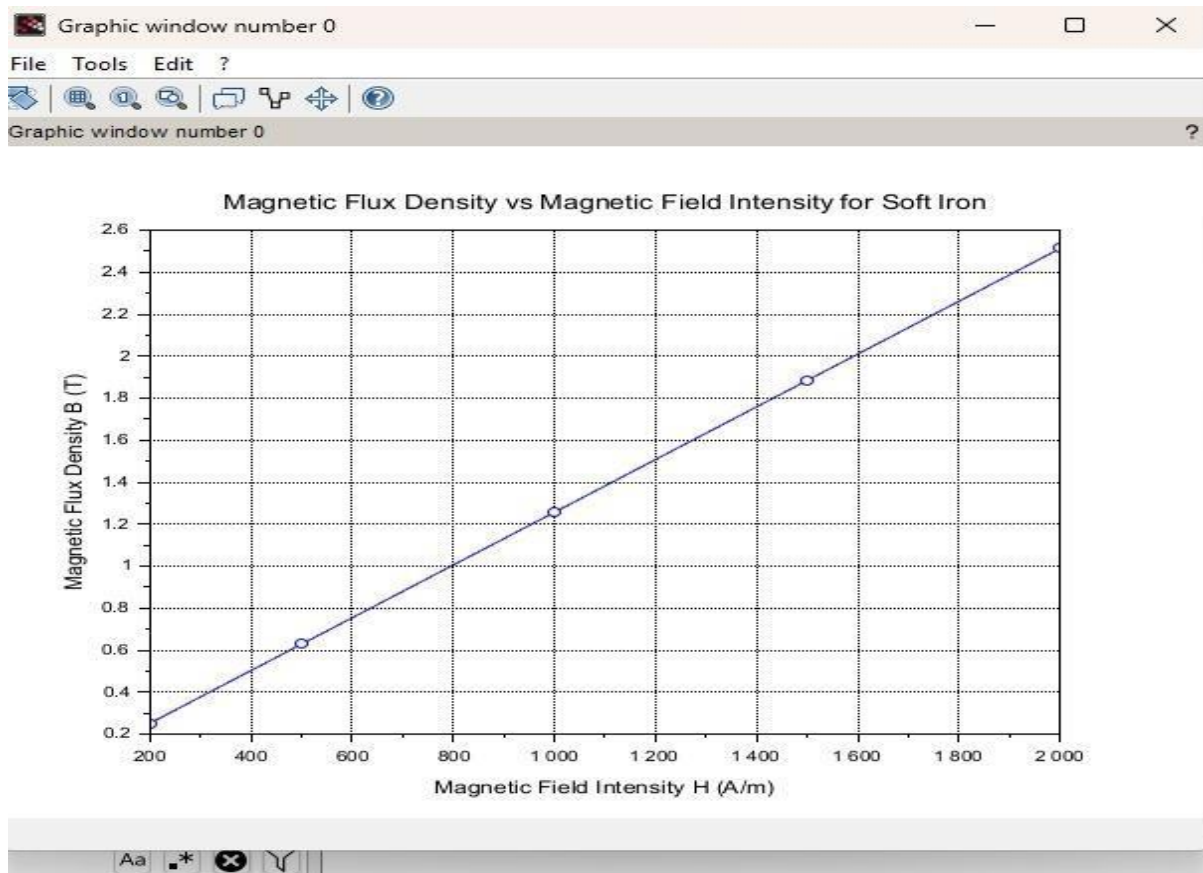
Magnetic Field Intensity (A/m)	Magnetic Flux Density (T)
200	0.251327
500	0.628319
1000	1.256637
1500	1.884956
2000	2.513274

The Variable Browser window is also open, showing the values of the variables used in the script:

Name	Value	Type	Visibility	Memory
B_values	1x5	Double	local	248 B
H	2e+03	Double	local	216 B
H_values	1x5	Double	local	248 B
L	0.1	Double	local	216 B
W	0.05	Double	local	216 B
i	5	Double	local	216 B
mu	0.00126	Double	local	216 B
mu_0	1.256e-06	Double	local	216 B
mu_r	1e+03	Double	local	216 B

The Command History window is also open, showing the commands executed in the console:

```
1 clear;
2 clear;
3 clf();
4
5 // Define constants
6 L = 0.1; // Length in meters (10 cm)
7 W = 0.05; // Width in meters (5 cm)
8
9 H_values = [200, 500, 1000, 1500, 2000]; // H in A/m
10
11 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
12 mu_r = 1000; // Relative permeability of soft iron
13
14 // Initialize array for magnetic flux density
15 B_values = zeros(1, length(H_values));
16
17 // Calculate B for each H value
18 for i = 1:length(H_values)
19     H = H_values(i);
20     mu = mu_0 * mu_r; // Absolute permeability
21     B_values(i) = mu * H; // B = mu * H
22 end
23
24 // Display results
25 disp('Magnetic Field Intensity (A/m) --- Magnetic Flux Density (T)');
26
27 for i = 1:length(H_values)
28     fprintf('%10.0f-----%.6f\n', H_values(i), B_values(i));
29 end
30
31 // Plot the results
32 plot(H_values, B_values, '-o');
33
34 xlabel('Magnetic Field Intensity H (A/m)');
35 ylabel('Magnetic Flux Density B (T)');
36 title('Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron');
37
38 grid();
39
```



Result : Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.